GCSE PHYSICS

Higher Tier – Paper 1 style

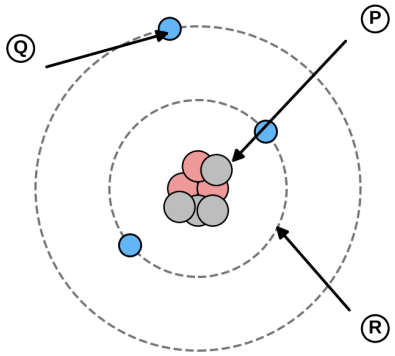
Topic 4.4: Atomic structure

Practice questions in the style of AQA GCSE Physics (8463). Every question is original; the layout, command words, mark allocations and mark scheme follow the real exam format.

Time allowed	About 140 minutes (roughly 1 minute per mark)
Materials	For this paper you must have: a ruler, a pencil and a calculator.
Instructions	• Use black ink or a black ball-point pen (pencil for diagrams and graphs). • Answer **all** questions. • Answer the questions in the spaces provided. • Do all rough work on a separate sheet. • Cross through any work you do not want to be marked.
Information	• The maximum mark for this paper is **133**. • The marks for questions are shown in brackets. • You are allowed to use a calculator. • You are reminded of the need for good English and clear presentation in your answers. • Questions that ask you to *describe* or *explain* are marked for the science you include, so give each point its own sentence.

Name: _____ Date: _____ Mark: _____ / 133

0 1 The diagram shows a model of an atom of lithium. A lithium atom has 3 protons and 4 neutrons.



0 1 . 1 Name the parts labelled **P**, **Q** and **R**. **[3 marks]**

0 1 . 2 State the relative charge and relative mass of a proton, a neutron and an electron. **[6 marks]**

0 1 . 3 Write down the atomic number and the mass number of this lithium atom. **[2 marks]**

0 1 . 4 Explain why an atom has no overall charge. **[2 marks]**

0 1 . 5 The radius of an atom is about 1×10^{-10} m. The radius of its nucleus is about 1×10^{-14} m. How many times larger is the radius of the atom than the radius of the nucleus? **[2 marks]**

0 1 . 6 Describe what happens to an atom when it absorbs electromagnetic radiation. **[2 marks]**

0 2 Isotopes of an element have the same number of protons but different numbers of neutrons.

0 2 . 1 Complete the table for neutral atoms. **[6 marks]**

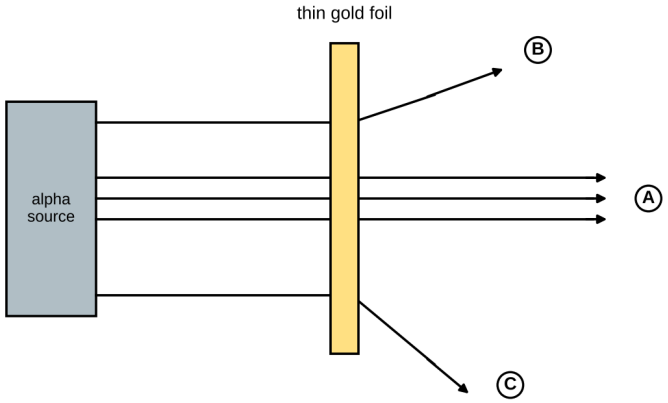
Atom	Protons	Neutrons	Electrons
carbon-12	6	6	6
carbon-14			
sodium-23			

0 2 . 2 Explain why carbon-12 and carbon-14 are isotopes. **[2 marks]**

0 2 . 3 A carbon atom loses an electron. State what the atom becomes and its overall charge. **[2 marks]**

0 2 . 4 Explain why the isotopes of an element have the same chemical properties. **[2 marks]**

0 3 The model of the atom has changed over time.



0 3 . 1 Describe the 'plum pudding' model of the atom. **[2 marks]**

0 3 . 2 The diagram shows the paths of alpha particles fired at thin gold foil in the alpha-scattering experiment. Which path, **A**, **B** or **C**, was followed by most of the alpha particles? **[1 mark]**

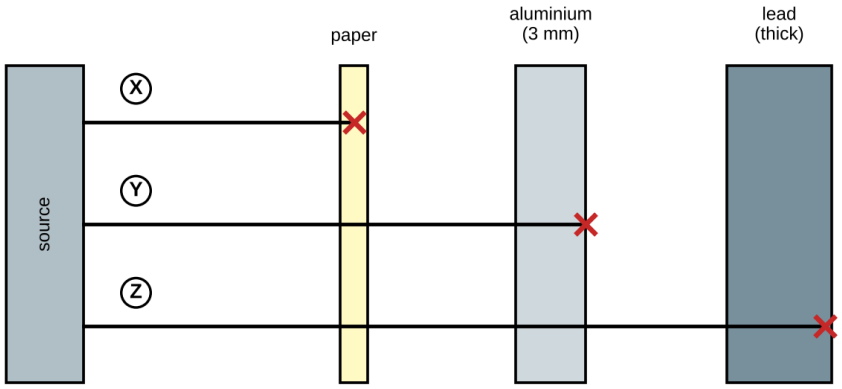
0 3 . 3 Explain what the paths of the alpha particles tell scientists about the structure of the atom. **[4 marks]**

0 3 . 4 Niels Bohr suggested a change to the nuclear model. Describe the change. **[2 marks]**

0 3 . 5 James Chadwick's experiments provided evidence for a new particle in the nucleus. Name the particle. **[1 mark]**

0 3 . 6 Suggest why the plum pudding model was accepted at first, but then replaced. **[2 marks]**

0 4 This question is about the properties of nuclear radiation.



0 4 . 1 Complete the table. **[6 marks]**

Radiation	What it is	What stops it
Alpha		
Beta		
Gamma		

0 4 . 2 The diagram shows three types of radiation **X**, **Y** and **Z** from a source passing through different materials. Identify **X**, **Y** and **Z**. **[3 marks]**

0 4 . 3 Which type of radiation is the most ionising? **[1 mark]**

0 4 . 4 Explain why alpha radiation is more dangerous than gamma radiation if the source is inside the body, but gamma is more dangerous if the source is outside. **[4 marks]**

0 5 Radioactive nuclei are unstable and emit radiation to become more stable.

$^{226}_{88}Ra \rightarrow {}^{?}_{?}Rn + {}^{4}_{2}He$ (alpha decay)

$^{14}_{6}C \rightarrow {}^{14}_{?}N + {}^{?}_{-1}e$ (beta decay)

0 5 . 1 Describe what happens to the mass number and atomic number of a nucleus when it emits an alpha particle. **[2 marks]**

0 5 . 2 Complete the nuclear equation for the alpha decay of radium-226 by writing the missing numbers in the equation. **[2 marks]**

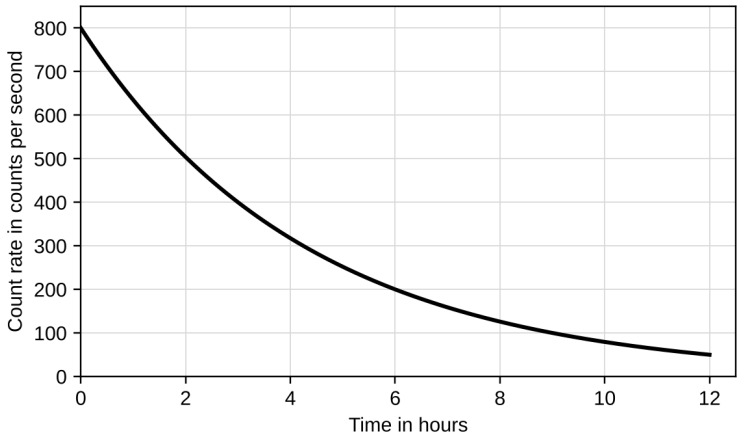
0 5 . 3 Complete the nuclear equation for the beta decay of carbon-14 by writing the missing numbers. **[2 marks]**

0 5 . 4 Describe what happens in the nucleus during beta decay. **[2 marks]**

0 5 . 5 Explain why the mass number does not change in beta decay. **[1 mark]**

0 5 . 6 Gamma radiation is often emitted after alpha or beta decay. Explain what effect gamma emission has on the nucleus. **[2 marks]**

0 6 A radioactive isotope has a count rate of 800 counts per second at the start of an experiment. The graph shows how the count rate changes with time.



0 6 . 1 What is meant by the **half-life** of a radioactive isotope? **[2 marks]**

0 6 . 2 Use the graph to find the half-life of this isotope. **[2 marks]**

0 6 . 3 Calculate the count rate after 12 hours. **[2 marks]**

0 6 . 4 Calculate the fraction of the original unstable nuclei remaining after 9 hours. **[2 marks]**

0 6 . 5 Another isotope has a half-life of 5 days. A sample has a count rate of 640 counts per second. Calculate the count rate after 15 days. **[3 marks]**

Answer = _____ counts/s

0 6 . 6 Calculate the net decline, as a percentage, in the activity of the isotope in the graph after 6 hours. **[3 marks]**

Answer = _____ %

0 6 . 7 Explain why it is impossible to predict when one particular nucleus will decay. **[2 marks]**

0 7 A student measures the radiation from a source with a Geiger–Müller tube. She first measures the background count rate for 5 minutes and records 60 counts. She then places a source close to the tube and records the count rate with different absorbers between source and tube.

0 7 . 1 Calculate the background count rate in counts per minute. **[2 marks]**

0 7 . 2 Give **two** sources of background radiation. **[2 marks]**

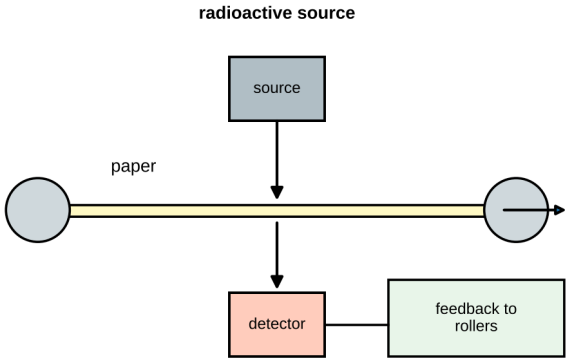
0 7 . 3 Complete the table by calculating the corrected count rate. **[3 marks]**

Absorber	**Measured count rate in counts per minute**	**Corrected count rate**
No absorber	540	
Paper	535	
3 mm aluminium	30	

0 7 . 4 Identify the type of radiation emitted by the source. Give a reason. **[2 marks]**

0 7 . 5 Explain why a source of gamma radiation would not give this pattern. **[2 marks]**

0 8 Radioactive isotopes have many uses but there are risks.



0 8 . 1 Describe the difference between **irradiation** and **contamination**. **[2 marks]**

0 8 . 2 Give **two** precautions to reduce the risks to people working with radioactive sources. **[2 marks]**

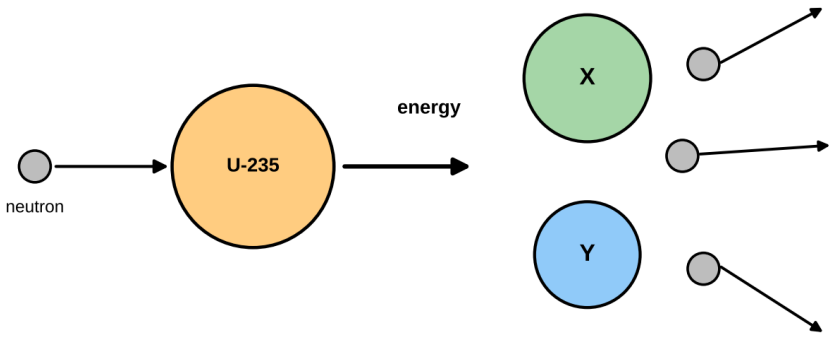
0 8 . 3 The diagram shows how the thickness of paper is monitored in a paper mill. Which type of radiation is used, alpha, beta or gamma? Give **two** reasons. **[3 marks]**

0 8 . 4 The detector shows a decrease in count rate. Explain what happens to the paper thickness. **[2 marks]**

0 8 . 5 A hospital uses a gamma-emitting isotope with a half-life of 6 hours as a tracer in the body. Explain why this is suitable. **[3 marks]**

0 8 . 6 Smoke alarms contain americium-241, an alpha source with a long half-life. Explain why alpha is suitable and why a long half-life matters. **[3 marks]**

0 9 This question is about nuclear fission and nuclear fusion.



0 9 . 1 Describe what happens in nuclear **fission**. **[2 marks]**

0 9 . 2 Complete the diagram labels: what are particles **X** and **Y**? **[2 marks]**

0 9 . 3 Describe how a chain reaction happens. **[3 marks]**

0 9 . 4 In a nuclear reactor, control rods are lowered into the core. Explain why. **[2 marks]**

0 9 . 5 Describe what happens in nuclear **fusion**. **[2 marks]**

0 9 . 6 Explain why fusion needs very high temperatures and pressures. **[2 marks]**

1 0 Read the question and answer in full sentences.

1 0 . 1 *Alpha, beta and gamma radiation can all be emitted by radioactive nuclei. Compare the three types of radiation. Include the nature, penetration and ionising power of each, and give one use for each. **[6 marks]**

1 0 . 2 *A hospital has a choice of two radioactive isotopes for treating a tumour deep inside the body: isotope A emits alpha radiation (half-life 3 days) and isotope B emits gamma radiation (half-life 6 hours). Evaluate which isotope is better for (i) treating from outside the body and (ii) being injected into the tumour. **[6 marks]**

END OF QUESTIONS

MARK SCHEME – Topic 4.4: Atomic structure

How to use this mark scheme: mark your own work in a different colour. A semicolon (;) separates marking points and each is worth one mark unless stated. A slash (/) separates alternative answers that are equally acceptable. Words in brackets are not needed to gain the mark. **Allow** = acceptable; **Ignore** = neither credited nor penalised; **Do not accept** = contradicts or spoils the answer. Marks are for the *science*, not the wording.

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.1	P: nucleus ; Q: electron ; R: electron shell / energy level ;	Allow 'orbit' for shell. One mark each.	3	AO1 4.4.1.1
01.2	proton: +1 ; mass 1 ; neutron: 0 ; mass 1 ; electron: −1 ; mass very small (1/1836 or negligible) ;		6	AO1 4.4.1.1
01.3	atomic number = 3 ; mass number = 7 ;		2	AO2 4.4.1.2
01.4	number of protons = number of electrons ; positive and negative charges cancel ;		2	AO2 4.4.1.1
01.5	$1 \times 10^{-10} \div 1 \times 10^{-14}$; 10 000 (times) ;		2	AO2 4.4.1.1
01.6	an electron moves to a higher energy level ; further from the nucleus ;		2	AO1 4.4.1.3
02.1	carbon-14: protons 6 ; neutrons 8 ; electrons 6 ; sodium-23: neutrons 12 ; electrons 11 ; protons 11 ;	Carbon-12 row is given. One mark for each correct box (max 6).	6	AO2 4.4.1.2
02.2	same number of protons (6) / same atomic number ; different number of neutrons (different mass number) ;		2	AO2 4.4.1.2
02.3	a (positive) ion ; +1 ;		2	AO2 4.4.1.3
02.4	chemical properties depend on the electrons / outer electrons ; isotopes have the same number of electrons (same protons) ;		2	AO2 4.4.1.2
03.1	a ball of positive charge ; with negative electrons embedded (spread) throughout ;		2	AO1 4.4.1.4
03.2	A		1	AO3 4.4.1.4

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.3	most pass straight through so most of the atom is empty space ; some deflected so the nucleus has a positive charge (repels positive alpha) ; a few bounce back (path C) so the nucleus is very small and dense / contains most of the mass ; this led to the nuclear model (replacing the plum pudding model) ;		4	AO3 4.4.1.4
03.4	electrons orbit the nucleus ; at specific distances / energy levels (shells) ;		2	AO1 4.4.1.4
03.5	neutron		1	AO1 4.4.1.4
03.6	it explained the evidence available at the time ; new experimental evidence (alpha scattering) did not fit the model ;		2	AO3 4.4.1.4
04.1	alpha: 2 protons and 2 neutrons / helium nucleus ; stopped by paper / few cm of air ; beta: fast-moving electron ; stopped by a few mm of aluminium ; gamma: electromagnetic wave ; reduced by thick lead / concrete ;		6	AO1 4.4.2.1
04.2	X: alpha ; Y: beta ; Z: gamma ;		3	AO3 4.4.2.1
04.3	alpha		1	AO1 4.4.2.1
04.4	alpha is strongly ionising ; and is absorbed by cells in the body (damages tissue) ; gamma is highly penetrating so can reach internal organs from outside ; alpha is stopped by skin / air from outside ;		4	AO3 4.4.2.1
05.1	mass number decreases by 4 ; atomic number decreases by 2 ;		2	AO1 4.4.2.3
05.2	mass number 222 ; atomic number 86 ;	$^{226}_{88}Ra \rightarrow {}^{222}_{86}Rn + {}^{4}_{2}He$	2	AO2 4.4.2.3
05.3	atomic number 7 ; mass number 0 ;	$^{14}_{6}C \rightarrow {}^{14}_{7}N + {}^{0}_{-1}e$	2	AO2 4.4.2.3
05.4	a neutron changes into a proton ; and an electron (beta particle) is emitted ;		2	AO1 4.4.2.3
05.5	the number of nucleons stays the same / neutron becomes proton		1	AO2 4.4.2.3
05.6	it loses excess energy ; no change in the number of protons or neutrons ;		2	AO2 4.4.2.3

Question	Answers	Extra information	Mark	AO / Spec. Ref.
06.1	the time taken for the number of (unstable) nuclei to halve ; or the time for the count rate / activity to fall to half ;		2	AO1 4.4.2.4
06.2	count rate = 400 ; 3.0 (hours) ;		2	AO3 4.4.2.4
06.3	12 ÷ 3 = 4 half-lives ; 800 → 400 → 200 → 100 → 50 (counts per second) ;		2	AO2 4.4.2.4
06.4	3 half-lives ; 1/8 ;		2	AO2 4.4.2.4
06.5	15 ÷ 5 = 3 half-lives ; 640 → 320 → 160 → 80 ; 80 (counts per second) ;		3	AO2 4.4.2.4
06.6	after 6 hours count rate = 200 ; decline = (800 − 200) ÷ 800 ; 75 (%) ;		3	AO3 4.4.2.4
06.7	decay is random ; can only predict the probability / the proportion that decays in a given time ;		2	AO2 4.4.2.2
07.1	60 ÷ 5 ; 12 (counts per minute) ;		2	AO2 4.4.2.1
07.2	radon gas / rocks ; cosmic rays ; food and drink ; medical (X-rays) ; nuclear power / weapons testing ;	Any two.	2	AO1 4.4.2.1
07.3	no absorber: 528 ; paper: 523 ; aluminium: 18 ;	Subtract the background of 12 counts per minute.	3	AO2 4.4.2.1
07.4	beta ; not stopped by paper but stopped by aluminium (nearly) ;		2	AO3 4.4.2.1
07.5	gamma would be only slightly reduced by aluminium ; (it needs thick lead / concrete to be reduced) ;		2	AO3 4.4.2.1
08.1	irradiation: exposure to radiation from a source outside the body ; contamination: radioactive atoms get on / into the body (and continue to irradiate) ;		2	AO1 4.4.2.5
08.2	use tongs / handle at a distance ; limit the time of exposure ; wear protective clothing / lead apron ; store sources in lead-lined containers ;	Any two.	2	AO1 4.4.2.5
08.3	beta ; alpha would be stopped by the paper / not penetrate ; gamma would pass through without much absorption (change not detectable) ;		3	AO3 4.4.2.5

Question	Answers	Extra information	Mark	AO / Spec. Ref.
08.4	paper is getting thicker (more beta absorbed) ; so the rollers are moved closer together ;		2	AO3 4.4.2.5
08.5	gamma passes out of the body and can be detected ; half-life long enough to complete the scan ; short enough that radiation does not stay in the body / patient exposure is low ;		3	AO3 4.4.2.5
08.6	alpha is easily absorbed by smoke so the detector senses the drop ; it is stopped by the casing so safe in the home ; long half-life means it doesn't need replacing often ;		3	AO3 4.4.2.5
09.1	a large unstable nucleus (e.g. uranium-235) absorbs a neutron and splits ; into two smaller nuclei, emitting neutrons and energy ;		2	AO1 4.4.4.1
09.2	two smaller (daughter) nuclei ; e.g. barium and krypton ;		2	AO2 4.4.4.1
09.3	neutrons released by fission ; are absorbed by other uranium nuclei ; causing those to split, releasing more neutrons (repeating) ;		3	AO1 4.4.4.1
09.4	they absorb neutrons ; to control the rate of the chain reaction / prevent runaway reaction ;		2	AO1 4.4.4.1
09.5	two light nuclei join together to form a heavier nucleus ; some mass is converted to energy released ;		2	AO1 4.4.4.2
09.6	nuclei are positively charged so repel ; need high kinetic energy to get close enough to fuse ;		2	AO2 4.4.4.2

Question	Answers	Extra information	Mark	AO / Spec. Ref.
10.1	**Level 3 (5–6 marks)**: A clear, detailed and logically ordered answer that covers all of the key areas, with correct scientific reasoning linking the ideas together. **Level 2 (3–4 marks)**: Some of the key areas are covered with some correct reasoning, but the answer lacks detail, balance or logical order in places. **Level 1 (1–2 marks)**: Simple, separate statements with little or no linking or reasoning. **0 marks**: No relevant content. **Indicative content** (credit any other correct science): • Level 3 (5–6): all three radiations compared with correct nature, penetration, ionisation and a use for each. • Level 2 (3–4): comparison of at least two with some correct detail. • Level 1 (1–2): simple statements. • Indicative content: alpha – helium nucleus, stopped by paper, strongly ionising, smoke alarms; beta – high-speed electron, stopped by few mm aluminium, moderately ionising, thickness gauge; gamma – electromagnetic radiation, penetrating (thick lead), weakly ionising, sterilising / tracers / radiotherapy.		6	AO1 4.4.2
10.2	**Level 3 (5–6 marks)**: A clear, detailed and logically ordered answer that covers all of the key areas, with correct scientific reasoning linking the ideas together. **Level 2 (3–4 marks)**: Some of the key areas are covered with some correct reasoning, but the answer lacks detail, balance or logical order in places. **Level 1 (1–2 marks)**: Simple, separate statements with little or no linking or reasoning. **0 marks**: No relevant content. **Indicative content** (credit any other correct science): • Level 3 (5–6): balanced evaluation of both isotopes in both situations with a justified choice. • Level 2 (3–4): some comparison. • Level 1 (1–2): simple statements. • Indicative content: from outside – gamma needed because alpha cannot penetrate; gamma short half-life reduces exposure but the dose must be controlled; injected – alpha is strongly ionising and would damage tumour cells locally with little damage elsewhere; longer half-life stays active longer but contamination risk; gamma would irradiate healthy tissue; safety precautions.		6	AO3 4.4.2.5

Where marks are lost in Topic 4.4

Use this page after marking. Tick the ones that cost you marks, then redo those questions closed-book.

- ☐ **Learn the numbers:** mass number (top) = protons + neutrons; atomic number (bottom) = protons. Neutrons = mass number − atomic number.
- ☐ **Nuclear equations:** the top numbers and bottom numbers must balance on both sides. Alpha: A − 4, Z − 2. Beta: A same, Z + 1.
- ☐ **Half-life:** draw a quick table (0, 1, 2, 3 half-lives) and halve each time. Read the half-life from the graph at **half** the starting value.
- ☐ **Background radiation** must be subtracted from every reading before you compare count rates.
- ☐ **Alpha vs gamma:** alpha is most ionising but least penetrating; gamma is least ionising and most penetrating. Inside the body alpha is the most dangerous.
- ☐ **Irradiation** = exposed from outside; **contamination** = radioactive material gets in or on you. Do not mix them up.
- ☐ **Evidence for the nuclear model:** most alpha particles straight through → empty space; some deflected → positive nucleus; few back → small, dense nucleus.
- ☐ **Fission** splits large nuclei (chain reaction, controlled by control rods). **Fusion** joins light nuclei (needs very high temperature to overcome repulsion).